# Advancing Confidence Calibration and Quantification in Medication Recommendation

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## Abstract

Medication recommendation (MR) has undergone rapid advancement in recent years, driven by its significant practical implications in healthcare. However, such high-risk scenarios still experience two critical yet overlooked challenges: the prevalent overconfidence in raw confidence for individual medications and the lack of a robust solution for confidence quantification in medication combinations. This paper represents the first in-depth study addressing this gap. We introduce two innovative methodologies tailored to the unique challenges of MR scenarios: 1) A discernible binning-based calibration method with theoretical guarantees for the confidence of individual medication. It guarantees distinct accuracy levels between adjacent bins and maintains consistent statistical reliability across calibration and test data, enabling calibrated confidence to reflect the correctness of medication recommendations distinctively. 2) A sample-based quantification method for the set confidence of medication combination, which is applicable for various existing performance metrics in MR. Utilizing representative deep MR models as backbones and conducting extensive experiments on the widely recognized MIMIC datasets, we empirically prove the effectiveness and robustness of our proposed methods. Our approaches not only improve the reliability of MR but also pave the way for more informed decision-making in clinical settings.

## CCS Concepts

• **Applied computing** → **Health care information systems**.

## Keywords

Medication Recommendation, Confidence Calibration, Histogram Binning, Conformal Risk Control

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## 1 Introduction

Medication recommendation (MR) systems have been receiving increasing attention in the field of machine learning. These systems are gaining significant interest in both industry and academia. Deep MR models [3, 16, 27, 29, 33] have been developed to suggest optimal medication combinations for patients. Despite considerable progress in improving their predictive accuracy, few works have focused on guaranteeing safety, which is of the utmost importance in clinical settings [31, 35].

To guarantee the safety of the recommendation, it is essential to provide users with reliable confidence. The probability outputted by a model typically serves as the raw confidence for user reference, and it is vital to minimize the discrepancy between the accuracy and the confidence of the predictions [9]. Addressing this issue requires confidence calibration, which has been studied in various areas [19, 23, 32, 37] but rarely received attention in existing MR research. Our work aims to conduct the first in-depth study on confidence calibration in MR, focusing on providing accurate and reliable confidence measures. Specifically, MR is a typical multi-label prediction task, where patient features from Electronic Health Records (EHRs) are the inputs, and the predicted medication combinations are the outputs. Therefore, we approach confidence calibration in MR at two levels: individual labels within the recommended set and the entire recommended set, as depicted in the rightmost part of Figure 2. We refer to these two levels as *individual confidence* and *set confidence*, respectively.

For individual confidence, as shown in Figure 1, our analysis uncovers a notable but overlooked issue of over-confidence in representative deep MR models. This trend indicates that the raw confidence assigned to the recommended medications is inordinately high, potentially compromising their reliability as the reference for users. These unreliable confidences necessitate confidence calibration, which can update the confidence value and provide the users with a safer reference. Apart from the over-confidence issue, the non-negligible incorrect recommendations within the recommended set pose another potential risk for users. As we observed,

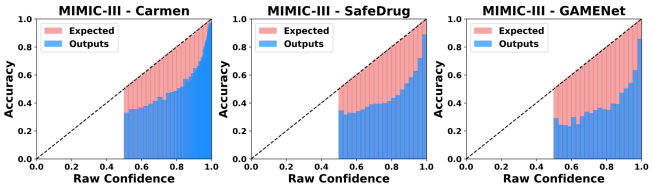



**Figure 1: Reliability diagrams of three MR models (Carmen, SafeDrug, and GAMENet) on MIMIC-III. The blue area denotes the actual accuracy of each model, while the red area signifies the expected accuracy. The average accuracy in each bin is consistently below the average confidence, indicating a prevalent issue of overconfidence in their uncalibrated confidences.**

recommendations from three representative MR models show that over 20% of medications are incorrectly recommended within the recommended set. This highlights the need for more discernible confidence to distinguish these incorrect recommendations. Prevalent calibration methods, like temperature scaling and regularization-based approaches [8], align confidences with original outcomes, keeping incorrect recommendations' confidences above 0.5. To address this, we adapt Histogram Binning (HB) [10] for individual confidence calibration. HB typically sets calibrated confidence based on the empirical accuracy of bins in held-out calibration data. However, conventional HB methods, such as fixed-width and uniform-mass binning heuristically split bins, thus failing to reflect natural data groupings. Furthermore, the accuracies of the adjacent bins may be too close, limiting effectiveness in MR. In summary, for individual confidence, we aim to achieve the discernible confidence calibration, which can 1) ensure that the calibration confidences of different bins are distinctive; and 2) assign the correct and incorrect recommendations with high and low confidence, respectively.

Another challenge for individual confidence calibration is the sparsity of medical data. Training deep models requires diverse labeled data, but labeled medical data is scarce, often only annotated at the image level [18]. This limits data-driven approaches for MR. Therefore, we consider the scenario where the size of calibration data is limited, ensuring our binning scheme maintains consistent statistical reliability across calibration and test data.

For set confidence, we also address the challenge of confidence quantification in MR, where the conventional definition of set accuracy is not applicable because traditionally a label set is *correct* only if it precisely matches the ground-truth label set. This definition is not practical in MR due to the typically large number of labels involved. Specifically, in the MIMIC-III ICU dataset [11], an average of around 19 medications are prescribed per visit, far exceeding the label count in the standard multi-label image and text datasets (*e.g.*, around 3 in the MSCOCO dataset [14]). As the number of labels increases, the probability of an exact match decreases, making the set accuracy metric approaching zero and thus misleading. Therefore, researchers in MR have adopted alternative metrics, such as Jaccard similarity, to evaluate the recommendation performance at the set level. However, until now, there has been a lack of an appropriate set confidence definition for MR scenarios. To address this gap, we propose a novel method for quantifying the confidence of the recommended set. This method is specifically designed for complex multi-label prediction scenarios like MR and is universally applicable to any performance metric, offering a versatile tool for set confidence quantification.

With the recommended sets outputted by the MR backbone and our proposed confidence calibration and quantification methods, the pipeline is shown in Figure 2. Our contributions are as follows:

- We identify and address the issues of overconfidence and the absence of medication combination confidence quantification in MR. These issues pose severe potential risks for the users and remain unsolved. To the best of our knowledge, we are the first to tackle these issues.
- We propose a sophisticated binning-based approach for individual confidence calibration with theoretical guarantees. Our approach ARC ensures distinct accuracy levels between adjacent bins and maintains consistent statistical reliability across calibration and test data, achieving discernible calibration in MR.
- We propose a novel method to quantify set confidence for various performance metrics of MR models.
- Through experiments with representative MR backbones and EHR datasets, we validate the efficacy and robustness of our methods, aiding more reliable and trustworthy decision-making for users in MR.

## 2 Related Work

### 2.1 Medication Recommendation

Using deep learning techniques for medication recommendation (MR) significantly enhances clinical efficiency and simplifies patient treatment processes. This field garners considerable attention in both industrial applications and academic research. MR involves suggesting a combination of medications based on a patient's physical condition. In the commonly used dataset MIMIC-III ICU [11], the average size of such medication combinations is 19, indicating a multi-label classification task with a relatively high number of labels.

The field of MR has witnessed remarkable advancement in consideration of various perspectives, including patient representation, medication representation, and external knowledge integration. For patient representation, the instance-based models [20, 22, 36] focus exclusively on a patient's current visit information; and the longitudinal-based models [5, 25, 28] incorporate a patient's historical treatment data. For medication representation, recent methodologies, including SafeDrug [29], MoleRec [30], and ACDNet [16], have increasingly employed graph neural networks (GNNs) to leverage information on molecular structures. Furthermore, GAMENet [21] and MEGACare [27] employed medical ontology knowledge bases to address the challenge of sparse medical data, and some works [4, 24, 26] integrated drug-drug interaction (DDI) knowledge bases to enhance the safety of the recommended medication combinations. The significant advancement in the efficacy of MR models highlights the necessity for further research into their reliability, which is particularly important due to the high-risk nature of the medical field.

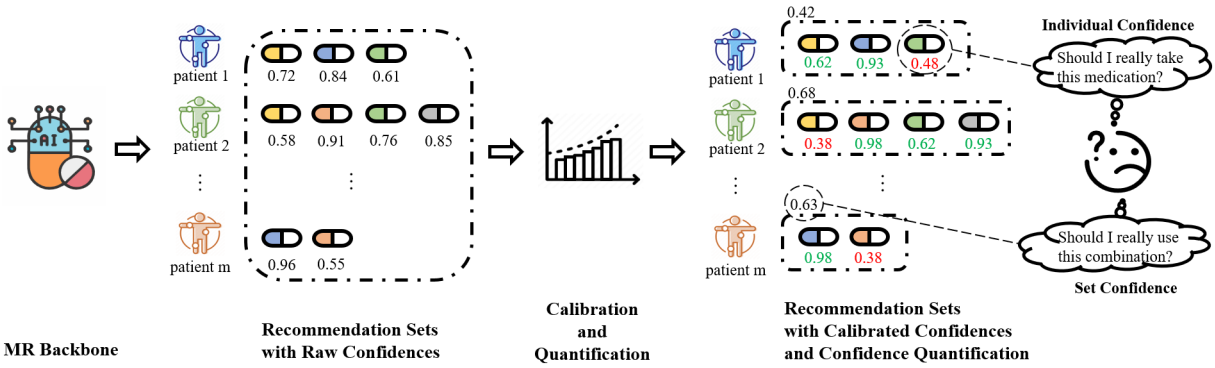



**Figure 2: The pipeline of confidence calibration and quantification. For individual confidence, our calibration method can group the confidences of the same level and effectively distinguish the correct and incorrect recommendations. For set confidence, we propose a quantification method that can provide the confidence for each recommended set.**

## 2.2 Multi-label Confidence Calibration

For set confidence, defined as the product of probabilities in multiple binary classifications, BR-rerank [13] utilizes gradient-boosted trees to reorder probabilities for multi-label image classification. In multi-label text classification, transformed into multi-class classifications, LP-ICP [15] provides reliable confidence measures based on inductive conformal prediction. Apart from the methods focusing on calibrating the confidence of the entire recommended set, in multi-label speech emotion recognition [6], class-balanced and emotion co-occurrence weight penalty loss, and class-specific temperature scaling for class-level calibration are used. Similarly, CASENT [7] in ultra-fine entity typing considers class occurrence frequency and devises class-specific platt scaling.

This work presents the first tailored confidence calibration for the MR scenario. Distinguishing from previous multi-label calibration work that focuses merely on either individual confidence or set confidence, our work uniquely addresses both levels concurrently, offering a comprehensive solution to confidence calibration in MR.

# 3 Preliminaries

## 3.1 Medication Recommendation

The Electronic Health Records (EHRs) comprise multiple EHR visits, and $\mathbf{x}_n \in \mathbb{R}^{1\times d}$ denotes the feature of the $n$-th visit. MR backbone can be viewed as a trained multi-label classifier $f$, and the output $\mathbf{c}_n = f(\mathbf{x}_n) = [c_{n1}, c_{n2}, \ldots, c_{nk}, \ldots, c_{nK}]$ is a probability vector. The scalar $c_{nk}$ is the raw confidence, denoting the probability that the $k$-th medication should be prescribed during the $n$-th visit. $K$ is the total number of medication categories. For the $n$-th visit, medications with a raw confidence $c_{nk} \geq 0.5$ are recommended as part of the medication combination $\hat{Y}_n$ outputted by the MR backbone.

## 3.2 Individual Confidence Calibration

Individual Confidence Calibration aims to refine raw confidence values into calibrated confidences, making them a better approximation of correctness. Post-hoc calibration leverages a calibration dataset $D_{cal} = (\mathbf{x}_n, \mathbf{c}_n, Y_n)_{n=1}^{|D_{cal}|}$, where $Y_n \subseteq \{1, 2, \ldots, K\}$ is the ground-truth label set and $\hat{Y}_n \subseteq \{1, 2, \ldots, K\}$ is the recommended set. Individual-level calibration focuses solely on the confidence levels of each medication, regardless of EHR visit indices. This implies that the actual calibration data is denoted as $D_{cal}^{med} = (c_n^{med}, Y_n^{med})_{n=1}^{|D_{cal}|*K}$, where $c_n^{med}$ and $Y_n^{med}$ are scalar values, corresponding to the raw confidence in recommending a single medication and its ground-truth label, respectively.

## 3.3 Conformal Risk Control

Conformal Risk Control (CRC) [2] is a technique derived from conformal prediction (CP) [1, 17], originally introduced for classification. CP is a distribution-free method used to construct prediction sets $C(X_{n+1})$ that provide coverage at a specific level for a test sample $X_{n+1}$ with true label $Y_{n+1}$, and can be formulated as $P(Y_{n+1} \notin C(X_{n+1})) \leq \alpha$, where $\alpha$ denotes the guaranteed statistical coverage (the controled risk). CRC generalizes CP by extending the simple inclusion criterion to an expected loss framework. Specifically, it ensures $\mathbb{E}[l(C(X_{n+1}), Y_{n+1}] \leq \alpha$ , where $l$ is a loss function measuring the discrepancy between the prediction set and the true label. When the loss $l$ is monotonic (e.g., the FNR), this technique allows us to control the risk associated with predictions. The theoretical foundation of CRC is based on the continuity and boundedness of the loss function. Proving the lower bound involves defining a jump function to establish the continuity properties required for the analysis. We refer the interested reader to [2] for more details on CRC and [1] for more details on CP.

# 4 Methodology

In this section, we first explain the method for discernible confidence calibration of individual labels within the recommended set. Thereafter, we introduce a novel set confidence quantification method specifically tailored for MR.

## 4.1 Discernible Individual Confidence Calibration

To facilitate a better understanding of our proposed method, we first briefly introduce the implementation of binning. Given the calibration data $D_{cal}^{med} = (c_n^{med}, Y_n^{med})_{n=1}^{|D_{cal}|*K}$, the test data $D_{test}^{med} =$

$(c_m^{med}, Y_m^{med})_{m=1}^{|D_{test}|*K}$, and the number of bins $B$, the process of a conventional binning scheme based calibration has four steps:

(1) Arrange all items in $D_{cal}^{med}$ based on the descending order of $c_n^{med}$.
(2) Assign each item to its corresponding bin using a predefined binning scheme.
(3) Calculate the accuracy $\{acc_i\}_{i=1}^{B}$ for each bin.
(4) For the test data, set the calibrated confidence $\hat{c}_m^{med}$ to $acc_i$ if the raw confidence $c_m^{med}$ falls within bin $b_i$.

Note that the key difference among the various binning schemes lies in the determination of the bin edges, *i.e.*, how the calibration data are partitioned (step 2). Compared to fixed-width and uniform-mass binning, which split the calibration data by the value of the raw confidence and the number of samples, respectively, our accuracy-aware binning split the calibration data by the False Negative Rate (FNR) value. In the following content, we will explain how our accuracy-aware binning can guarantee the distinctiveness of the accuracy. Note that in our analysis, although we ignore the indices of the EHR visits and merge all the medications recommended per visit, the medications recommended from different visits are regarded as distinct because of their different raw confidences, even if they have the same label, achieving the patient-specific calibration.

*4.1.1 Accuracy-Aware Binning.* We denote the cumulative predicted labels in the first $i$ bins as $\hat{Y}_{<i}$, thus we have $\hat{Y}_{<i} = \bigcup_{s=1}^{i} b_s$ and $\hat{Y}_{<B} = \hat{Y}$, where $b_s$ is the $s$-th bin, and $B$ is the total number of bins and $\hat{Y}$ is the recommended set. We define $\{\alpha_i\}_{i=1}^{B}$ as a sequence of predetermined FNR values, and $\alpha_i$ represents the FNR value of $\hat{Y}_{<i}$. Specifically, the generation of prediction set $\hat{Y}_{<i}$ can be defined by a function $C(\lambda_i)$, *i.e.*,

$$\hat{Y}_{<i} = C(\lambda_i) = \{n : c_n^{med} \geq 1 - \lambda_i\}, \tag{1}$$

where $1 - \lambda_i$ is the bin edge of bin $b_i$.

The essence of our binning scheme is in identifying appropriate bin edges $1 - \lambda_i$ that satisfies $\text{FNR}(\hat{Y}_{<i} = C(\lambda_i), Y) = \alpha_i$. Let $Y$ be the set of ground-truth labels, the numerical relationship between FNR value and $\hat{Y}_{<i}$ can be written as:

$$\text{FNR}(\hat{Y}_{<i}, Y) = 1 - \frac{\left|Y \cap \hat{Y}_{<i}\right|}{|Y|}. \tag{2}$$

The accuracy $acc_i$ of bin $b_i$ is calculated as:

$$acc_i = \frac{\left|Y \cap (\hat{Y}_{<i+1} - \hat{Y}_{<i})\right|}{\left|(\hat{Y}_{<i+1} - \hat{Y}_{<i})\right|}. \tag{3}$$

Given that both $\alpha_i$ and $\alpha_{i+1}$ are predetermined FNR values, *i.e.*, $\text{FNR}(\hat{Y}_{<i}, Y)$ and $\text{FNR}(\hat{Y}_{<i+1}, Y)$, their corresponding $\hat{Y}_{<i}$ and $\hat{Y}_{<i+1}$ can be generated with Equation (1) and (2). Intuitively, the value of $acc_i$ (Equation (3)) is governed by adjusting the values of $\alpha_i$, indicating that our binning scheme is accuracy-aware. Theoretically, our accuracy-aware binning scheme can ensure a theoretical lower bound for the accuracy differences between adjacent bins as formulated in the proposition below.

Proposition 4.1. *Given a backbone and the accuracy-aware binning scheme, there exits $acc_i - acc_{i+1} \geq 0$. Furthermore, assume that $\left|\hat{Y}_{<i}\right| = \beta_i |Y|$, then $acc_i - acc_{i+1}$ has a lower bound of*

$$\frac{(1 - \alpha_{i+2} - \beta_{i+1})\alpha_i + (1 - \alpha_i - \beta_{i+2})\alpha_{i+1} + (1 - \alpha_{i+1} - \beta_i)\alpha_{i+2}}{(\beta_{i+1} - \beta_i)(\beta_{i+2} - \beta_{i+1})}.$$

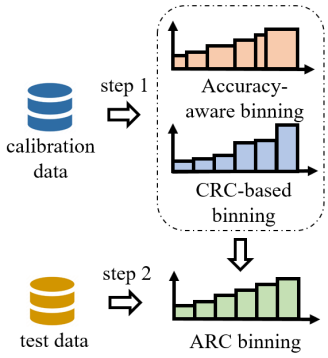



**Figure 3: Two steps of discernible individual confidence calibration. We propose the accuracy-aware binning scheme and the CRC-based binning scheme and apply them to calibration data respectively. We combine the two proposed schemes as our ARC binning scheme for test data.**

The proof is available in the Appendix. With the lower bound of the accuracy differences between adjacent bins, *i.e.*, $acc_i - acc_{i+1}$, our accuracy-aware binning scheme can ensure the numerical diversity of $\{acc_i\}_{i=1}^{B}$. The calibrated confidence can be discernible based on the theoretically guaranteed distinctive accuracies. Another property is that the calibrated confidences are robust due to the theoretical guarantee that $acc_i \geq acc_{i+1}$. We will further verify this property in the experiments.

Our accuracy-aware binning scheme allows computing a series of bins that satisfy $\{\text{FNR}(\hat{Y}_{<i} = C(\lambda_i), Y) = \alpha_i\}_{i=1}^{B}$. Note that up to this stage, we have established bin edges $\{1 - \lambda_i\}_{i=1}^{B}$ and their corresponding accuracies $\{acc_i\}_{i=1}^{B}$ by the predetermined FNR values $\{\alpha_i\}_{i=1}^{B}$. Traditional binning methods typically directly assign these bin accuracies as calibrated confidence to the test data partitioned by these bin edges. However, we argue that these bin edges should not be directly applied to segment test data due to potential disparities between the calibration and test datasets, especially in MR where the calibration data is limited.

Recall that the accuracy-aware binning split the calibration data by a series of predetermined FNR values $\{\alpha_i\}_{i=1}^{B}$, considering the potential disparities between the calibration and test data, we propose to update $\{\alpha_i\}_{i=1}^{B}$, thereby applying the new bin edges to split test data. Therefore, we propose to apply conformal risk control (CRC) [2] to disturb the predetermined FNR values $\{\alpha_i\}_{i=1}^{B}$ to $\{\alpha_i'\}_{i=1}^{B}$, aiming to acquire the more proper bin edges $\{1 - \lambda_i'\}_{i=1}^{B}$ to split the test data. In the next section, we will explain how CRC works to maintain the statistical consistency of the FNR values between calibration and test data.

*4.1.2 Conformal Risk Control Based Binning.* We assume that the bin count for calibration and test data is the same. To ensure similar accuracies between bins of identical indices in the test and calibration datasets, we utilize Conformal Risk Control (CRC) to update

$\{\alpha_i\}_{i=1}^{B}$ from the previous section, approximately aligning their accuracies. CRC is a technique that provides statistical coverage guarantees for the prediction set $\hat{Y}_{<j}$ of samples in test data (indices $i$ and $j$ denote the bin indices in calibration and test data, respectively). Specifically, for any test sample, when selecting a prediction set $\hat{Y}_{<j}$ with CRC, the following relationship can be maintained with ideal conditions:

$$\dot{\alpha}_j - \frac{2}{|D_{cal}|+1} \leq \mathbb{E}[l(\hat{Y}_{<j}, Y')] \leq \dot{\alpha}_j, \tag{4}$$

for any bounded loss function $l$ that decreases as the size of $\hat{Y}_{<j}$ increases. $\dot{\alpha}_j$ is the desired statistical guarantee for the prediction set $\hat{Y}_{<j}$ in test data. In our study, we set $l(\hat{Y}_{<j}, Y') = \text{FNR}(\hat{Y}_{<j}, Y')$.

Ideally, the FNR values of $\hat{Y}_{<j} = \bigcup_{s=1}^{j} b_s$ on the test data should follow the pattern $\{\text{FNR}(\hat{Y}_{<j}, Y') = \alpha_i\}_{i=j=1}^{B}$, where $Y'$ is the ground-truth label set of the test data.

As our objective is to have $\text{FNR}(\hat{Y}_{<j}, Y') = \alpha_i, s.t., i = j$, which cannot be directly attained due to the unknown $Y'$. However, CRC, as a post-hoc method, can bound $\mathbb{E}[\text{FNR}(\hat{Y}_{<j}, Y')]$ solely utilizing calibration data $D_{cal}$ as shown in Equation (4). By setting $\dot{\alpha}_j = \alpha_i$, we can approximate $\mathbb{E}[\text{FNR}(\hat{Y}_{<j}, Y')] = \alpha_i' \approx \alpha_i$.

Therefore, we apply CRC to calibration data. To bound the value of the FNR, the CRC finds $\lambda_i'$ that satisfies

$$\frac{|D_{cal}|}{|D_{cal}|+1}\text{FNR}(\hat{Y}_{<i} = C(\lambda_i'), Y) + \frac{1}{|D_{cal}|+1} = \alpha_i. \tag{5}$$

That is,

$$\text{FNR}(\hat{Y}_{<i} = C(\lambda_i'), Y) = \alpha_i' = \frac{|D_{cal}|+1}{|D_{cal}|}(\alpha_i - \frac{1}{|D_{cal}|+1}), \tag{6}$$

where $\alpha_i'$ can be regarded as the disturbed $\alpha_i$. With $\lambda_i'$ generated by $\alpha_i'$ with Equation (1) and (2), we can adjust the bin edges for partitioning test data to $\{1-\lambda_i'\}_{i=1}^{B}$, ensuring the statistical guarantees of $\mathbb{E}[\text{FNR}(\hat{Y}_{<j}, Y')] \approx \alpha_i$ on test data. To summarize, the CRC-based binning scheme provides a bounded risk for binning test data with FNR values similar to those of the calibration data.

*4.1.3 Accuracy-aware Risk Control Binning.* As shown in Figure 3, our discernible individual confidence calibration consists of two steps: 1) Using the calibration data, we facilitate an accuracy-aware binning scheme for discernible reference and tailor the bin edges by CRC-based binning scheme. 2) For the test data, combine the two binning schemes from Step 1 as our new binning scheme, which is referred to as Accuracy-aware Risk Control (ARC) binning.

ARC binning leverages the accuracy-aware binning scheme to derive a series of discernible accuracies $\{acc_i\}_{i=1}^{B}$ based on the predetermined FNR values, and sets the bin edges as $\{1-\lambda_i'\}_{i=1}^{B}$ to guarantee the alignment of FNR values by CRC-based binning. Overall, ARC binning considers the accuracy within each bin and the statistical reliability of data, allowing us to obtain discernible calibrated confidences and appropriately assign these confidence values to each test sample.

## 4.2 Sample-based Set Confidence Quantification

Generally, *accuracy* is a commonly used metric to evaluate the performance of a model and is the target that calibrated confidences need to align to. However, as we have discussed before, this metric is limited in MR as it is almost impossible to recommend the set that exactly matches the ground-truth label set. Therefore, among the existing works, alternative metrics such as Jaccard similarity have been widely applied to evaluate the recommended medication sets in MR. Likewise, as a complex multi-label prediction task, MR requires innovative solutions to address the challenge of set confidence quantification, which is essential for the reliable evaluation of recommended medication combinations. However, there have been no appropriate solutions for defining set confidence in such scenarios. Therefore, we propose a comprehensive quantification approach to calculate the confidence of these metrics.

The key idea of our approach is to leverage the individually calibrated confidences assigned to each label, quantifying the confidence of a metric. This quantification is then capable of evaluating the performance of a recommended set, such as the confidence quantification of the Jaccard similarity.

For the $m$-th EHR visit $(x_m, c_m, Y_m)$ from the test data, let the vector $\hat{c}_m = [\hat{c}_{m1}, \hat{c}_{m2}, \ldots, \hat{c}_{mK}]$ represents the calibrated confidence after individual confidence calibration, where $K$ is the number of medication categories. The calibrated confidence for each medication $\hat{c}_{mk}$ is determined by:

$$\hat{c}_{mk} = \begin{cases} h(c_{mk}), \ k \in \hat{Y}_m, \\ \beta, \ k \notin \hat{Y}_m, \end{cases} \tag{7}$$

where $h(c_{mk})$ is the calibration scheme applied to the model's raw confidence when the $k$-th medication belongs to the recommended set for the $m$-th EHR visit. For medications not included in the recommended set, a fixed confidence $\beta$ is assigned, which is derived from calibration data and represents the average frequency of labels that were incorrectly omitted from the recommendations.

Then, we employ a Bernoulli sampling technique to generate $\ddot{Y}_m$, i.e.,

$$\ddot{Y}_m \sim \text{Bernoulli}(\hat{c}_m). \tag{8}$$

$\ddot{Y}_m$ serves as an approximation of $Y_m$ from the test data. Then, for any metric $J$ applicable to the backbone and computed with the ground-truth label set $Y_m$, we estimate it by

$$J(f(x_m), Y_m) \approx J(f(x_m), \ddot{Y}_m), \tag{9}$$

where $f$ is the backbone. Specifically, for metrics, such as Jaccard similarity, that take the recommended set $\hat{Y}_m$ as input, we have $J(\hat{Y}_m, Y_m) \approx J(\hat{Y}_m, \ddot{Y}_m)$.

# 5 Experiments

We experimented extensively on various representative deep MR backbones to evaluate the efficacy of our calibration method at the individual level and the quantification method at the set confidence level. The backbones tested include Carmen [4], SafeDrug [29], and GAMENet [21]. Our evaluations utilized two widely recognized EHR datasets, MIMIC-III [11] and MIMIC-IV [12].

## 5.1 Experimental Setting

*5.1.1 Evaluation Metrics (Individual).* We employed the following metrics to assess the performance of individual label confidence calibration.

- ***ECE*** (Expected Calibration Error) evaluates the alignment between bin-wise confidence and accuracy. With $M = |D_{test}| *$

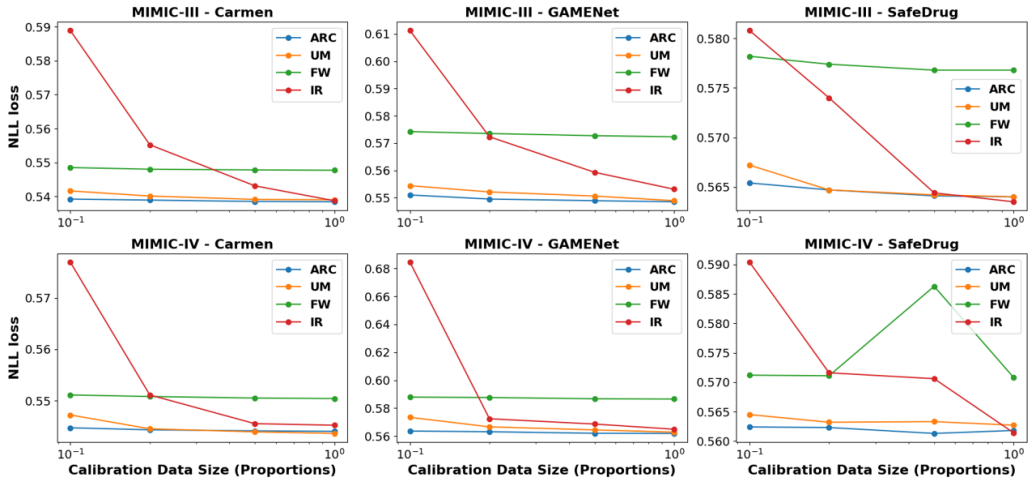



**Figure 4: The NLL loss values on test data with calibration data of different proportional sizes. Our proposed binning scheme, ARC (blue line), consistently shows the best performance and is the most robust one.**

$K$ individual labels and $B$ bins, ***ECE*** is defined as

$$\boldsymbol{ECE} = \sum_{i=1}^{B} \frac{|b_i|}{M} \left| acc(b_i) - conf(b_i) \right|,$$

where $b_i$ denotes the subset of individual labels in the $i$-th bin, $acc(b_i)$ and $conf(b_i)$ are the accuracy and confidence of $b_i$, respectively.

- ***ACC***$^+$ denotes the proportion of correct predictions calculated by the calibrated confidence $\hat{c}_m^{med}$. Here, correct predictions represent the correct recommendations with $\hat{c}_m^{med} \geq 0.5$ and the incorrect recommendations with $\hat{c}_m^{med} < 0.5$. This metric aims to evaluate the ability of calibrated confidence to distinguish between correct and incorrect recommendations.
- ***NLL*** (Negative Log-Likelihood) loss quantifies the discrepancy between confidence and ground truth. ***NLL*** loss, defined as

$$\boldsymbol{NLL} = -\sum_{m=1}^{M} [y_m log(\hat{c}_m^{med}) + (1 - y_m) log(1 - \hat{c}_m^{med})],$$

reaches its optimum when correct recommendations have higher confidence and incorrect ones have lower confidence. $\hat{c}_m^{med}$ is the calibrated confidence of the $m$-th medication and $y_m$ is the corresponding ground-truth label (equals to 0 or 1). ***NLL*** loss serves as a measure of the performance of the discernible confidence calibration and can be regarded as the synthesis of the ***ECE*** and ***ACC***$^+$.

*5.1.2 Evaluation Metrics (Set).* We employed three common metrics in MR: **Jaccard** similarity,**F1** score, and Precision-Recall AUC (**PRAUC**). We then measured the differences between the ground-truth values of these metrics and our set confidence estimations using both Mean Squared Error (***MSE***) and ***ECE***.

*5.1.3 Baselines.* We compared the proposed **ARC** (Accuracy-aware Risk Control) with the following prevalent post-hoc calibration approaches: **Temperature Scaling (TS)** [9] scales logits with a scalar global temperature. **Vector Scaling (VS)** is applied in **SER** [6]. It uses the temperature in vector form and introduces classwise biases. **Ensemble Temperature Scaling (ETS)** [34] learns the ensemble of the uncalibrated, TS-calibrated, and uniform probabilistic outputs to scale logits. **Platt Scaling (PS)** is employed in **CASENET** [7] and fits a logistic regression model into the raw confidences generated by the backbone. **Isotonic Regression (IR)** is utilized in **BR-rerank** [13] by fitting an isotonic function to the raw confidences. **Fixed-Width binning (FW)** uniformly partitions the probability interval [0, 1] to determine bin edges. **Uniform-Mass binning (UM)** determines bin edges by uniformly distributing the number of samples across bins. Results from the uncalibrated backbones (**Uncal**) are also presented for comparison.

*5.1.4 Implementation Details.* To ensure comprehensive evaluations, we initially split the dataset into training, calibration, and test sets with a 4:1:1 ratio. All datasets were randomly shuffled five times with different seeds. For binning schemes, we set bin counts $B$ of 10, 20, and 40, and calculated ECE with 25 bins while evaluating the performance of individual confidence calibration. In all experiments, the FNR value for each bin was predetermined as $\{\alpha_i = (B - i + 1) * \alpha/B\}_{i=1}^{B}$, where $\alpha$ is the FNR value of the recommended medications calculated over the calibration dataset. For set confidence quantification, we performed 50 rounds of Bernoulli sampling and used 40 bins for the evaluation of ECE.

## 5.2 Individual Confidence Calibration

Figure 4 depicts the ***NLL*** loss on test data when utilizing calibration data of varying proportional sizes. It can be observed that our ARC binning consistently outperforms other methods across various datasets and MR backbones. Though UM is on par with ARC when the size of calibration data is large enough, its performance is not robust with the shrink of the calibration data size. In contrast, FW shows robustness, but cannot effectively reduce the ***NLL*** loss. Note that the size of calibration data is crucial in healthcare-related tasks, where labeled data is typically scarce and expensive. ARC reliably

**Table 1: Performance comparison of individual confidence calibration in terms of ECE, ACC$^+$, and NLL loss values (mean ± std).**

| MR Model | Method | MIMIC-III | | | MIMIC-IV | | |
|---|---|---|---|---|---|---|---|
| | | ECE (↓, %) | ACC$^+$ (↑, %) | NLL (↓) | ECE (↓, %) | ACC$^+$ (↑, %) | NLL (↓) |
| Carmen | Uncal | $20.11_{\pm 0.07}$ | $65.88_{\pm 0.16}$ | $0.7216_{\pm 0.0028}$ | $20.00_{\pm 0.20}$ | $64.36_{\pm 0.22}$ | $0.7047_{\pm 0.0062}$ |
| | TS | $13.77_{\pm 0.12}$ | $65.88_{\pm 0.16}$ | $0.6020_{\pm 0.0033}$ | $13.97_{\pm 0.27}$ | $64.36_{\pm 0.22}$ | $0.6053_{\pm 0.0048}$ |
| | SER (VS) | $12.47_{\pm 0.11}$ | $65.88_{\pm 0.16}$ | $0.5912_{\pm 0.0045}$ | $13.67_{\pm 0.22}$ | $64.36_{\pm 0.22}$ | $0.6055_{\pm 0.0053}$ |
| | ETS | $13.76_{\pm 0.14}$ | $65.88_{\pm 0.16}$ | $0.6017_{\pm 0.0093}$ | $15.40_{\pm 0.17}$ | $64.36_{\pm 0.22}$ | $0.6203_{\pm 0.0036}$ |
| | CASENET (PS) | $8.86_{\pm 0.12}$ | $71.21_{\pm 0.11}$ | $0.5693_{\pm 0.0017}$ | $8.28_{\pm 0.17}$ | $71.10_{\pm 0.18}$ | $0.5715_{\pm 0.0026}$ |
| | BR-Rank (IR) | $1.86_{\pm 0.12}$ | $71.26_{\pm 0.12}$ | $0.5756_{\pm 0.0360}$ | $1.89_{\pm 0.46}$ | $70.96_{\pm 0.20}$ | $0.5844_{\pm 0.0250}$ |
| | FW | $1.64_{\pm 0.39}$ | $71.11_{\pm 0.16}$ | $0.5486_{\pm 0.0017}$ | $1.55_{\pm 0.28}$ | $70.98_{\pm 0.34}$ | $0.5505_{\pm 0.0031}$ |
| | UM | $1.58_{\pm 0.28}$ | $71.33_{\pm 0.22}$ | $0.5455_{\pm 0.0021}$ | $1.52_{\pm 0.36}$ | $70.91_{\pm 0.18}$ | $0.5469_{\pm 0.0056}$ |
| | ARC | $\mathbf{1.38_{\pm 0.43}}$ | $\mathbf{71.34_{\pm 0.17}}$ | $\mathbf{0.5391_{\pm 0.0014}}$ | $\mathbf{1.30_{\pm 0.38}}$ | $\mathbf{71.11_{\pm 0.18}}$ | $\mathbf{0.5439_{\pm 0.0030}}$ |
| SafeDrug | Uncal | $24.78_{\pm 0.12}$ | $62.52_{\pm 0.13}$ | $0.8359_{\pm 0.0023}$ | $25.86_{\pm 0.22}$ | $57.78_{\pm 0.29}$ | $0.8089_{\pm 0.0056}$ |
| | TS | $14.76_{\pm 0.13}$ | $62.52_{\pm 0.13}$ | $0.6289_{\pm 0.0052}$ | $16.76_{\pm 0.30}$ | $57.78_{\pm 0.29}$ | $0.6402_{\pm 0.0064}$ |
| | SER (VS) | $14.27_{\pm 0.21}$ | $62.52_{\pm 0.13}$ | $0.6320_{\pm 0.0074}$ | $15.25_{\pm 0.18}$ | $57.78_{\pm 0.29}$ | $0.6499_{\pm 0.0044}$ |
| | ETS | $14.80_{\pm 0.26}$ | $62.52_{\pm 0.13}$ | $0.6291_{\pm 0.0024}$ | $19.76_{\pm 0.32}$ | $57.78_{\pm 0.29}$ | $0.6741_{\pm 0.0058}$ |
| | CASENET (PS) | $9.17_{\pm 0.22}$ | $69.00_{\pm 0.18}$ | $0.5977_{\pm 0.0017}$ | $8.66_{\pm 0.12}$ | $68.77_{\pm 0.25}$ | $0.5899_{\pm 0.0022}$ |
| | BR-Rank (IR) | $2.52_{\pm 0.34}$ | $69.41_{\pm 0.18}$ | $0.5830_{\pm 0.0085}$ | $1.82_{\pm 0.12}$ | $\mathbf{70.39_{\pm 0.13}}$ | $0.5881_{\pm 0.0139}$ |
| | FW | $2.23_{\pm 0.51}$ | $69.18_{\pm 0.53}$ | $0.5788_{\pm 0.0018}$ | $1.30_{\pm 0.27}$ | $70.31_{\pm 0.13}$ | $0.5707_{\pm 0.0005}$ |
| | UM | $2.22_{\pm 0.39}$ | $69.38_{\pm 0.16}$ | $0.5711_{\pm 0.0055}$ | $1.17_{\pm 0.25}$ | $70.22_{\pm 0.10}$ | $0.5626_{\pm 0.0025}$ |
| | ARC | $\mathbf{1.98_{\pm 0.30}}$ | $\mathbf{69.51_{\pm 0.16}}$ | $\mathbf{0.5662_{\pm 0.0019}}$ | $\mathbf{1.07_{\pm 0.29}}$ | $70.25_{\pm 0.08}$ | $\mathbf{0.5611_{\pm 0.0007}}$ |
| GAMENet | Uncal | $28.54_{\pm 0.32}$ | $59.57_{\pm 0.33}$ | $0.9296_{\pm 0.0023}$ | $32.24_{\pm 0.16}$ | $55.92_{\pm 0.32}$ | $1.0304_{\pm 0.0101}$ |
| | TS | $17.65_{\pm 0.23}$ | $59.57_{\pm 0.33}$ | $0.6439_{\pm 0.0028}$ | $19.08_{\pm 0.17}$ | $55.92_{\pm 0.32}$ | $0.6647_{\pm 0.0091}$ |
| | SER (VS) | $16.17_{\pm 0.21}$ | $59.57_{\pm 0.33}$ | $0.6427_{\pm 0.0021}$ | $17.46_{\pm 0.23}$ | $55.92_{\pm 0.32}$ | $0.6774_{\pm 0.0082}$ |
| | ETS | $17.80_{\pm 0.19}$ | $59.57_{\pm 0.33}$ | $0.6481_{\pm 0.0037}$ | $19.00_{\pm 0.15}$ | $55.92_{\pm 0.32}$ | $0.6643_{\pm 0.0092}$ |
| | CASENET (PS) | $10.65_{\pm 0.48}$ | $69.53_{\pm 0.27}$ | $0.5923_{\pm 0.0030}$ | $10.69_{\pm 0.20}$ | $68.77_{\pm 0.25}$ | $0.6037_{\pm 0.0042}$ |
| | BR-Rank (IR) | $2.70_{\pm 0.89}$ | $70.91_{\pm 0.13}$ | $0.6018_{\pm 0.0276}$ | $2.48_{\pm 0.33}$ | $70.69_{\pm 0.23}$ | $0.6353_{\pm 0.0620}$ |
| | FW | $2.50_{\pm 1.10}$ | $70.85_{\pm 0.27}$ | $0.5875_{\pm 0.0019}$ | $1.90_{\pm 0.71}$ | $70.48_{\pm 0.30}$ | $0.5878_{\pm 0.0019}$ |
| | UM | $2.60_{\pm 0.62}$ | $70.92_{\pm 0.24}$ | $0.5617_{\pm 0.0019}$ | $1.75_{\pm 0.41}$ | $70.74_{\pm 0.16}$ | $0.5692_{\pm 0.0058}$ |
| | ARC | $\mathbf{2.34_{\pm 0.68}}$ | $\mathbf{71.05_{\pm 0.27}}$ | $\mathbf{0.5521_{\pm 0.0024}}$ | $\mathbf{1.74_{\pm 0.34}}$ | $\mathbf{70.78_{\pm 0.29}}$ | $\mathbf{0.5636_{\pm 0.0018}}$ |

ensures discernible confidence calibration, which is paramount in clinical decision-making environments. Considering the sparsity of the medical data in practice, we fix the size of the calibration data and test data with a 1:10 ratio. Specifically, we have the following observations:

**Obs1: Calibration is crucial for MR. ARC offers a reliable calibration tool that aligns confidence with accuracy.** The ***ECE*** metric indicates how well the accuracy and the confidence are aligned. Table 1 reveals that the raw confidence produced by MR backbones is generally unreliable, resulting in high ***ECE*** (Uncal). Notably, IR, FW, UM, and our approach ARC achieve an ***ECE*** of approximately 0.01 or 0.02, indicating a significant improvement. Among these methods, ARC overall exhibits a slight advantage over others, demonstrating the effectiveness of our approach. PS falls short in ***ECE*** because the distribution of calibrated confidence is inconsistent with logistic regression. TS-based methods (TS, VS, and ETS) aim to align confidence within the recommended set by scaling the model's logits. This alignment restricts calibrated confidences to remain higher than 0.5, even for incorrect recommendations. This highlights the limitation of TS in multi-label classification tasks.

**Obs2: ARC can further improve the accuracy of the backbones.** A pivotal aspect of ARC is its capacity to augment the model accuracy of the backbones, denoted as ***ACC***$^+$ in Table 1. This improvement is achieved by reselecting the recommended set according to the calibrated confidence. All methods except TS-based ones contribute to improved performance of MR models in terms of ***ACC***$^+$, with ARC demonstrating a slight superiority. This reinforces ARC's proficiency in more accurately distinguishing correct from incorrect recommendations within the recommended set. In contrast, TS-based methods cannot contribute to ***ACC***$^+$ due to their incapability of reducing the confidence of incorrect recommendations, as previously discussed. The improvement in ***ACC***$^+$ highlights the significance of discernible confidence calibration for MR models, facilitating more precise assessments regarding individual medications within recommended combinations and improving the safety and reliability of the recommendations.

**Obs3: ARC achieves the most discernible individual confidence calibration.** Compared to ***ECE*** and ***ACC***$^+$, we consider ***NLL*** loss as a more discernible metric. A lower ***NLL*** loss implies that the model, on average, assigns higher confidences to correct recommendations and lower confidences to incorrect recommendations. Despite the comparable performance of IR, FW, UM, and ARC in terms of ***ECE*** and ***ACC***$^+$, Table 1 shows that ARC consistently outperforms others in terms of ***NLL*** loss. This is because ARC applies FNR values to split data to theoretically guarantee the relationship of the accuracies between the adjacent bins, and can maintain consistent statistical reliability across both calibration and test data. On the contrary, IR, FW, and UM fit the confidence distribution heuristically, regardless of the relationship between the

accuracies within different bins and the statistical reliability of data. Specifically, IR merely fits the isotonic functions and cannot effectively handle the out-of-distribution samples. FW and UM show inferior performance because high raw confidences are assigned to a majority of samples in the recommended set. TS-based methods, constrained by the binary classification threshold and the observed fact that the accuracy of the raw confidence is around 60% (Table 1, Uncal), tend to overlook incorrect recommendations, leading to the poorest performance.

**Obs4: ARC exhibits superior robustness compared to other methods.** We conducted experiments to assess the stability and reliability of the proposed method. Table 2 shows the results after we disturb the distribution of the data. For both the calibration data and test data, we set the labels of the samples with raw confidence over 0.99 to 0. Remarkably, ARC overall shows the best performance, especially in terms of $ACC^+$ and ***NLL***. It can be observed that ARC outperforms the runner-up method by 3% to 10% in terms of $ACC^+$. ARC also consistently achieves the lowest ***NLL*** loss and outperforms the runner-up method by 0.03 to 0.11. This superiority indicates that ARC can provide safe and reliable calibrated confidence even when the data distribution has been disturbed. IR is competitive with ARC in terms of ***ECE*** as our disturbance cannot destroy the monotonicity of the distribution. However, IR cannot guarantee the reliability of the recommended medications due to the low $ACC^+$ and ***NLL***. More analysis and comparison regarding robustness are available in the Appendix (Table 4).

**Obs5: Both accuracy-aware and CRC-based binning schemes reliably ensure the discernible confidence calibration.** We also conducted the ablation study, of which the results and the detailed analysis are available in the Appendix (Table 5).

## 5.3 Set Confidence Quantification

We conducted experiments to validate the effectiveness of our sample-based quantification method. Table 3 shows the results of quantifying recommended set confidence. The quantification based on discernible individual confidence calibration achieves a more accurate approximation than that of uncalibrated individual confidence. The findings validate the effectiveness of our quantification approach in approximating the true values of various model performance metrics at the set level.

Figure 5 visualizes the performance of the GAMENet backbone in terms of Metric Value for samples from the MIMIC-III dataset, alongside their corresponding confidence levels. The x-axis represents the value of set confidence quantification, while the y-axis corresponds to the model's performance measured by a specific metric. Ideally, well-calibrated set confidence quantifications should closely align with the diagonal line, indicating a concurrence between confidence levels and actual performance. Observing the right column of the figures, we find that the distribution of uncalibrated set confidence quantification deviates significantly from the diagonal line, showing a positive correlation with the metric value. This correlation provides indirect evidence supporting the validity of our quantification approach. ARC, as shown in the left column, aligns closely with the diagonal line across various metrics, indicating a substantial enhancement in the calibration of set confidence quantification. Moreover, our set confidence exhibits a more even

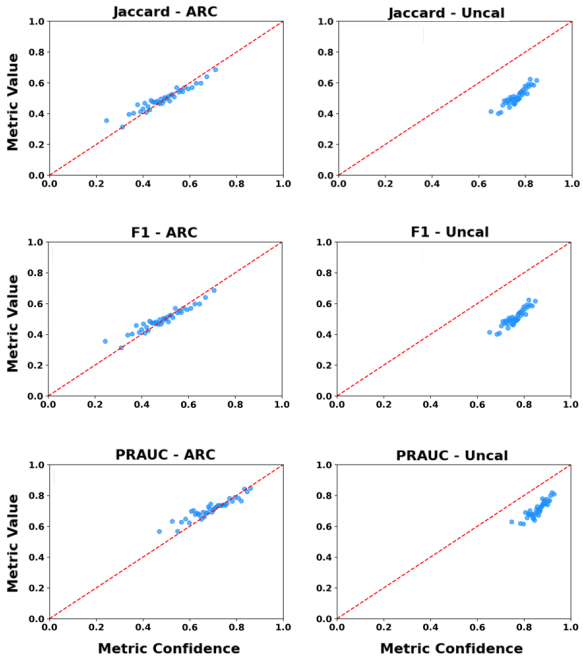



**Figure 5: Set confidence quantification w.r.t. various metrics on MIMIC-III dataset with GAMENet backbone. Each dot corresponds to a bin derived from a binning scheme.**

distribution along the diagonal, indicating that the quantification derived from discernible individual confidence is effective in facilitating discernible set confidence quantification. Consequently, the confidence calibration and set confidence quantification methods we propose enable users to more accurately assess the trustworthiness of recommended medication combinations. This alignment assures users that a higher confidence level reliably indicates a greater likelihood of correct recommendations, thereby facilitating more informed decision-making.

## 6 Conclusion and Discussion

In this paper, we have identified and addressed critical yet overlooked issues in deep learning-driven medication recommendation (MR) scenarios: the prevalent overconfidence in raw confidences for individual medications and the absence of confidence quantification for medication combinations. As the first in-depth research on confidence calibration in MR, we considered the need for user reference and the sparsity of medical data and proposed two innovative methods to address these challenges: the ARC binning scheme for discernible individual confidence calibration and a sample-based set confidence quantification method. Backed by both theoretical guarantees and empirical evidence, our methods effectively mitigate the overconfidence problem and introduce a reliable solution for quantifying confidence in medication combinations, thereby paving the way for more reliable decision-making in MR.

The scarcity and high cost of labeled medical data are the primary limitations of our calibration method. Due to the scarcity and expense of labeled data, our method may struggle in cases with

**Table 2: With the disturbed data, the performance comparison of robust individual confidence calibration in terms of ECE, ACC$^+$, and NLL loss values (mean ± std).**

| MR Model | Method | MIMIC-III | | | MIMIC-IV | | |
|---|---|---|---|---|---|---|---|
| | | ECE (↓, %) | ACC$^+$ (↑, %) | NLL (↓) | ECE (↓, %) | ACC$^+$ (↑, %) | NLL (↓) |
| Carmen | Uncal | $32.03_{\pm 0.63}$ | $50.98_{\pm 0.14}$ | $1.5476_{\pm 0.0087}$ | $30.72_{\pm 0.16}$ | $53.65_{\pm 0.20}$ | $1.2874_{\pm 0.0065}$ |
| | TS | $20.78_{\pm 0.12}$ | $50.98_{\pm 0.14}$ | $0.7282_{\pm 0.0093}$ | $19.32_{\pm 0.24}$ | $53.65_{\pm 0.20}$ | $0.7114_{\pm 0.0048}$ |
| | SER (VS) | $16.93_{\pm 0.14}$ | $50.98_{\pm 0.14}$ | $0.7455_{\pm 0.0075}$ | $15.40_{\pm 0.27}$ | $53.65_{\pm 0.20}$ | $0.7293_{\pm 0.0031}$ |
| | ETS | $20.83_{\pm 0.18}$ | $50.98_{\pm 0.14}$ | $0.7293_{\pm 0.0066}$ | $24.37_{\pm 0.11}$ | $53.65_{\pm 0.20}$ | $0.8531_{\pm 0.0052}$ |
| | CASENET (PS) | $23.08_{\pm 0.34}$ | $55.75_{\pm 0.30}$ | $0.7378_{\pm 0.0020}$ | $19.06_{\pm 1.02}$ | $60.07_{\pm 0.14}$ | $0.7156_{\pm 0.0007}$ |
| | BR-Rank (IR) | $1.77_{\pm 0.12}$ | $56.29_{\pm 0.26}$ | $0.7095_{\pm 0.0285}$ | $1.76_{\pm 0.75}$ | $60.23_{\pm 0.22}$ | $0.6930_{\pm 0.0241}$ |
| | FW | $1.84_{\pm 0.41}$ | $56.29_{\pm 0.26}$ | $0.6790_{\pm 0.0004}$ | $1.89_{\pm 0.73}$ | $60.31_{\pm 0.22}$ | $0.6659_{\pm 0.0012}$ |
| | UM | $5.92_{\pm 1.91}$ | $66.11_{\pm 5.28}$ | $0.6266_{\pm 0.0272}$ | $8.71_{\pm 1.05}$ | $60.22_{\pm 0.23}$ | $0.6961_{\pm 0.0263}$ |
| | ARC | $\mathbf{1.63_{\pm 0.55}}$ | $\mathbf{69.84_{\pm 1.13}}$ | $\mathbf{0.5693_{\pm 0.0203}}$ | $\mathbf{1.54_{\pm 0.65}}$ | $\mathbf{67.66_{\pm 0.50}}$ | $\mathbf{0.6027_{\pm 0.0056}}$ |
| SafeDrug | Uncal | $38.77_{\pm 0.25}$ | $48.53_{\pm 0.25}$ | $1.6483_{\pm 0.0091}$ | $35.86_{\pm 0.21}$ | $47.80_{\pm 0.23}$ | $1.3876_{\pm 0.0071}$ |
| | TS | $21.29_{\pm 0.17}$ | $48.53_{\pm 0.25}$ | $0.7451_{\pm 0.0074}$ | $21.52_{\pm 0.23}$ | $47.80_{\pm 0.23}$ | $0.7401_{\pm 0.0057}$ |
| | SER (VS) | $19.82_{\pm 0.21}$ | $48.53_{\pm 0.25}$ | $0.7450_{\pm 0.0043}$ | $18.52_{\pm 0.22}$ | $47.80_{\pm 0.23}$ | $0.7459_{\pm 0.0047}$ |
| | ETS | $21.39_{\pm 0.18}$ | $48.53_{\pm 0.25}$ | $0.7465_{\pm 0.0082}$ | $27.73_{\pm 0.29}$ | $47.80_{\pm 0.23}$ | $0.8639_{\pm 0.0080}$ |
| | CASENET (PS) | $19.63_{\pm 0.28}$ | $55.33_{\pm 0.34}$ | $0.7271_{\pm 0.0013}$ | $15.65_{\pm 0.47}$ | $60.29_{\pm 0.18}$ | $0.6960_{\pm 0.0016}$ |
| | BR-Rank (IR) | $\mathbf{1.78_{\pm 0.89}}$ | $55.61_{\pm 0.22}$ | $0.6860_{\pm 0.0057}$ | $1.46_{\pm 0.40}$ | $60.26_{\pm 0.11}$ | $0.6727_{\pm 0.0098}$ |
| | FW | $1.84_{\pm 0.65}$ | $55.54_{\pm 0.25}$ | $0.6810_{\pm 0.0008}$ | $1.54_{\pm 0.38}$ | $60.01_{\pm 0.30}$ | $0.6618_{\pm 0.0005}$ |
| | UM | $6.20_{\pm 1.60}$ | $64.65_{\pm 5.12}$ | $0.6409_{\pm 0.0200}$ | $9.28_{\pm 0.83}$ | $60.11_{\pm 0.12}$ | $0.7231_{\pm 0.0194}$ |
| | ARC | $4.84_{\pm 0.23}$ | $\mathbf{67.54_{\pm 0.76}}$ | $\mathbf{0.6173_{\pm 0.0130}}$ | $\mathbf{1.34_{\pm 0.50}}$ | $\mathbf{70.41_{\pm 0.41}}$ | $\mathbf{0.5538_{\pm 0.0092}}$ |
| GAMENet | Uncal | $50.07_{\pm 0.57}$ | $38.41_{\pm 0.50}$ | $2.5006_{\pm 0.0675}$ | $52.48_{\pm 0.33}$ | $51.06_{\pm 0.29}$ | $2.3017_{\pm 0.0059}$ |
| | TS | $26.26_{\pm 0.31}$ | $38.41_{\pm 0.50}$ | $0.7815_{\pm 0.0041}$ | $29.12_{\pm 0.24}$ | $51.06_{\pm 0.29}$ | $0.8110_{\pm 0.0061}$ |
| | SER (VS) | $28.06_{\pm 0.23}$ | $38.41_{\pm 0.50}$ | $0.8327_{\pm 0.0021}$ | $30.83_{\pm 0.17}$ | $51.06_{\pm 0.29}$ | $0.8510_{\pm 0.0027}$ |
| | ETS | $25.10_{\pm 0.26}$ | $38.41_{\pm 0.50}$ | $0.7607_{\pm 0.0059}$ | $28.09_{\pm 0.28}$ | $51.06_{\pm 0.29}$ | $0.7924_{\pm 0.0044}$ |
| | CASENET (PS) | $23.42_{\pm 0.18}$ | $44.96_{\pm 1.11}$ | $0.7085_{\pm 0.0038}$ | $22.17_{\pm 0.58}$ | $64.43_{\pm 0.29}$ | $0.6860_{\pm 0.0030}$ |
| | BR-Rank (IR) | $\mathbf{1.55_{\pm 0.61}}$ | $61.59_{\pm 0.50}$ | $0.6876_{\pm 0.0259}$ | $\mathbf{1.59_{\pm 0.86}}$ | $56.28_{\pm 0.29}$ | $0.6520_{\pm 0.0070}$ |
| | FW | $1.62_{\pm 0.56}$ | $62.33_{\pm 0.28}$ | $0.6577_{\pm 0.0020}$ | $2.01_{\pm 0.72}$ | $56.22_{\pm 0.14}$ | $0.6438_{\pm 0.0021}$ |
| | UM | $13.48_{\pm 3.19}$ | $54.04_{\pm 8.97}$ | $0.6993_{\pm 0.0640}$ | $10.34_{\pm 4.01}$ | $66.15_{\pm 6.79}$ | $0.6534_{\pm 0.0679}$ |
| | ARC | $2.58_{\pm 0.26}$ | $\mathbf{72.67_{\pm 0.32}}$ | $\mathbf{0.5089_{\pm 0.0114}}$ | $3.23_{\pm 0.80}$ | $\mathbf{69.76_{\pm 1.06}}$ | $\mathbf{0.5369_{\pm 0.0219}}$ |

**Table 3: Performance comparison of set confidence quantification (mean ± std).**

| MR Model | Method | MIMIC-III | | | | | | MIMIC-IV | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Jaccard | | PRAUC | | F1 score | | Jaccard | | PRAUC | | F1 score | |
| | | MSE (↓, %) | ECE (↓, %) | MSE (↓, %) | ECE (↓, %) | MSE (↓, %) | ECE (↓, %) | MSE (↓, %) | ECE (↓, %) | MSE (↓, %) | ECE (↓, %) | MSE (↓, %) | ECE (↓, %) |
| Carmen | Uncal | $4.04_{\pm 0.03}$ | $17.06_{\pm 0.09}$ | $2.47_{\pm 0.05}$ | $11.28_{\pm 0.16}$ | $2.73_{\pm 0.02}$ | $13.50_{\pm 0.05}$ | $4.03_{\pm 0.05}$ | $16.35_{\pm 0.22}$ | $2.79_{\pm 0.04}$ | $10.65_{\pm 0.28}$ | $2.91_{\pm 0.04}$ | $13.37_{\pm 0.16}$ |
| | ARC | $\mathbf{0.97_{\pm 0.07}}$ | $\mathbf{1.82_{\pm 0.11}}$ | $\mathbf{1.19_{\pm 0.02}}$ | $\mathbf{1.89_{\pm 0.10}}$ | $\mathbf{0.84_{\pm 0.02}}$ | $\mathbf{1.61_{\pm 0.09}}$ | $\mathbf{1.34_{\pm 0.02}}$ | $\mathbf{2.09_{\pm 0.10}}$ | $\mathbf{1.67_{\pm 0.06}}$ | $\mathbf{2.43_{\pm 0.18}}$ | $\mathbf{1.19_{\pm 0.02}}$ | $\mathbf{1.95_{\pm 0.11}}$ |
| SafeDrug | Uncal | $5.39_{\pm 0.05}$ | $20.64_{\pm 0.13}$ | $3.47_{\pm 0.03}$ | $14.43_{\pm 0.08}$ | $3.71_{\pm 0.04}$ | $16.57_{\pm 0.11}$ | $5.95_{\pm 0.05}$ | $21.76_{\pm 0.14}$ | $3.61_{\pm 0.03}$ | $13.85_{\pm 0.13}$ | $4.40_{\pm 0.03}$ | $18.07_{\pm 0.11}$ |
| | ARC | $\mathbf{1.02_{\pm 0.02}}$ | $\mathbf{1.78_{\pm 0.17}}$ | $\mathbf{1.32_{\pm 0.04}}$ | $\mathbf{1.92_{\pm 0.10}}$ | $\mathbf{0.93_{\pm 0.02}}$ | $\mathbf{1.74_{\pm 0.17}}$ | $\mathbf{1.22_{\pm 0.02}}$ | $\mathbf{1.51_{\pm 0.10}}$ | $\mathbf{1.62_{\pm 0.05}}$ | $\mathbf{2.10_{\pm 0.18}}$ | $\mathbf{1.18_{\pm 0.02}}$ | $\mathbf{1.51_{\pm 0.08}}$ |
| GAMENet | Uncal | $7.68_{\pm 0.22}$ | $25.44_{\pm 0.52}$ | $3.58_{\pm 0.10}$ | $14.98_{\pm 0.41}$ | $4.97_{\pm 0.14}$ | $19.79_{\pm 0.39}$ | $9.50_{\pm 0.22}$ | $28.30_{\pm 0.44}$ | $4.55_{\pm 0.14}$ | $16.74_{\pm 0.46}$ | $6.44_{\pm 0.12}$ | $22.54_{\pm 0.29}$ |
| | ARC | $\mathbf{1.09_{\pm 0.05}}$ | $\mathbf{2.34_{\pm 0.34}}$ | $\mathbf{1.55_{\pm 0.53}}$ | $\mathbf{2.64_{\pm 0.39}}$ | $\mathbf{0.94_{\pm 0.05}}$ | $\mathbf{2.24_{\pm 0.26}}$ | $\mathbf{1.34_{\pm 0.06}}$ | $\mathbf{2.44_{\pm 0.12}}$ | $\mathbf{1.70_{\pm 0.06}}$ | $\mathbf{3.05_{\pm 0.30}}$ | $\mathbf{1.22_{\pm 0.43}}$ | $\mathbf{2.40_{\pm 0.13}}$ |

insufficient calibration data. However, this challenge can be mitigated using conformal risk control (CRC), which adjusts bin edges to maintain applicability in most scenarios, even with limited data. While CRC helps with data scarcity, it may still face difficulties in extreme cases with very limited data.

Overall, our method offers clinicians quick, reliable confidence estimates and is non-parametric, making it time-efficient without requiring complex computations. It is robust, data-efficient, and well-suited for various practical applications, particularly in clinical settings with limited data.

In the future, we plan to incorporate patient-specific information for more personalized confidence calibration and expand its use to other healthcare contexts to enhance robustness and safety in clinical decision-making.

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## References


[1] Anastasios N Angelopoulos and Stephen Bates. 2021. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511* (2021).

[2] Anastasios N Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. 2022. Conformal risk control. *arXiv preprint arXiv:2208.02814* (2022).

[3] Suman Bhoi, Mong Li Lee, Wynne Hsu, and Ngiap Chuan Tan. 2024. REFINE: a fine-grained medication recommendation system using deep learning and personalized drug interaction modeling. *Advances in Neural Information Processing Systems* 36 (2024).

[4] Qianyu Chen, Xin Li, Kunnan Geng, and Mingzhong Wang. 2023. Context-aware safe medication recommendations with molecular graph and DDI graph embedding. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. 7053–7060.

[5] Edward Choi, Mohammad Taha Bahadori, Jimeng Sun, Joshua Kulas, Andy Schuetz, and Walter Stewart. 2016. Retain: An interpretable predictive model for healthcare using reverse time attention mechanism. *Advances in neural information processing systems* 29 (2016).

[6] Huang-Cheng Chou, Lucas Goncalves, Seong-Gyun Leem, Chi-Chun Lee, and Carlos Busso. 2023. The Importance of Calibration: Rethinking Confidence and Performance of Speech Multi-label Emotion Classifiers. *Interspeech 2023* (2023).

[7] Yanlin Feng, Adithya Pratapa, and David R Mortensen. 2023. Calibrated seq2seq models for efficient and generalizable ultra-fine entity typing. *arXiv preprint arXiv:2311.00835* (2023).

[8] Jakob Gawlikowski, Cedrique Rovile Njieutcheu Tassi, Mohsin Ali, Jongseok Lee, Matthias Humt, Jianxiang Feng, Anna Kruspe, Rudolph Triebel, Peter Jung, Ribana Roscher, et al. 2023. A survey of uncertainty in deep neural networks. *Artificial Intelligence Review* 56, Suppl 1 (2023), 1513–1589.

[9] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. 2017. On calibration of modern neural networks. In *International conference on machine learning*. PMLR, 1321–1330.

[10] Chirag Gupta and Aaditya Ramdas. 2021. Distribution-free calibration guarantees for histogram binning without sample splitting. In *International conference on machine learning*. PMLR, 3942–3952.

[11] Alistair EW Johnson, Tom J Pollard, Lu Shen, Li-wei H Lehman, Mengling Feng, Mohammad Ghassemi, Benjamin Moody, Peter Szolovits, Leo Anthony Celi, and Roger G Mark. 2016. MIMIC-III, a freely accessible critical care database. *Scientific data* 3, 1 (2016), 1–9.

[12] Alistair EW Johnson, David J Stone, Leo A Celi, and Tom J Pollard. 2018. The MIMIC Code Repository: enabling reproducibility in critical care research. *Journal of the American Medical Informatics Association* 25, 1 (2018), 32–39.

[13] Cheng Li, Virgil Pavlu, Javed Aslam, Bingyu Wang, and Kechen Qin. 2020. Learning to calibrate and rerank multi-label predictions. In *Machine Learning and Knowledge Discovery in Databases: European Conference, ECML PKDD 2019, Würzburg, Germany, September 16–20, 2019, Proceedings, Part III*. Springer, 220–236.

[14] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. 2014. Microsoft coco: Common objects in context. In *Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part V 13*. Springer, 740–755.

[15] Lysimachos Maltoudoglou, Andreas Paisios, Ladislav Lenc, Jiří Martínek, Pavel Král, and Harris Papadopoulos. 2022. Well-calibrated confidence measures for multi-label text classification with a large number of labels. *Pattern Recognition* 122 (2022), 108271.

[16] Jiacong Mi, Yi Zu, Zhuoyuan Wang, and Jieyue He. 2024. ACDNet: Attention-guided Collaborative Decision Network for effective medication recommendation. *Journal of Biomedical Informatics* 149 (2024), 104570.

[17] Harris Papadopoulos. 2008. Inductive conformal prediction: Theory and application to neural networks. In *Tools in artificial intelligence*. Citeseer.

[18] Appan K Pujitha and Jayanthi Sivaswamy. 2018. Solution to overcome the sparsity issue of annotated data in medical domain. *CAAI Transactions on Intelligence Technology* 3, 3 (2018), 153–160.

[19] Weizhu Qian, Yan Zhao, Dalin Zhang, Bowei Chen, Kai Zheng, and Xiaofang Zhou. 2023. Towards a Unified Understanding of Uncertainty Quantification in Traffic Flow Forecasting. *IEEE Transactions on Knowledge and Data Engineering* (2023).

[20] Junyuan Shang, Tengfei Ma, Cao Xiao, and Jimeng Sun. 2019. Pre-training of graph augmented transformers for medication recommendation. *arXiv preprint arXiv:1906.00346* (2019).

[21] Junyuan Shang, Cao Xiao, Tengfei Ma, Hongyan Li, and Jimeng Sun. 2019. Gamenet: Graph augmented memory networks for recommending medication combination. In *proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 33. 1126–1133.

[22] Yanchao Tan, Chengjun Kong, Leisheng Yu, Pan Li, Chaochao Chen, Xiaolin Zheng, Vicki S Hertzberg, and Carl Yang. 2022. 4sdrug: Symptom-based set-to-set small and safe drug recommendation. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 3970–3980.

[23] Jingyuan Wang, Ning Wu, Xinxi Lu, Wayne Xin Zhao, and Kai Feng. 2019. Deep trajectory recovery with fine-grained calibration using kalman filter. *IEEE Transactions on Knowledge and Data Engineering* 33, 3 (2019), 921–934.

[24] Shanshan Wang, Pengjie Ren, Zhumin Chen, Zhaochun Ren, Jun Ma, and Maarten de Rijke. 2019. Order-free medicine combination prediction with graph convolutional reinforcement learning. In *Proceedings of the 28th ACM international conference on information and knowledge management*. 1623–1632.

[25] Yanda Wang, Weitong Chen, Dechang Pi, Lin Yue, Sen Wang, and Miao Xu. 2021. Self-Supervised Adversarial Distribution Regularization for Medication Recommendation.. In *IJCAI*. 3134–3140.

[26] Jialun Wu, Yuxin Dong, Zeyu Gao, Tieliang Gong, and Chen Li. 2023. Dual attention and patient similarity network for drug recommendation. *Bioinformatics* 39, 1 (2023), btad003.

[27] Jialun Wu, Kai He, Rui Mao, Chen Li, and Erik Cambria. 2023. MEGACare: Knowledge-guided multi-view hypergraph predictive framework for healthcare. *Information Fusion* 100 (2023), 101939.

[28] Chaoqi Yang, Cao Xiao, Lucas Glass, and Jimeng Sun. 2021. Change matters: Medication change prediction with recurrent residual networks. *arXiv preprint arXiv:2105.01876* (2021).

[29] Chaoqi Yang, Cao Xiao, Fenglong Ma, Lucas Glass, and Jimeng Sun. 2021. Safedrug: Dual molecular graph encoders for recommending effective and safe drug combinations. *arXiv preprint arXiv:2105.02711* (2021).

[30] Nianzu Yang, Kaipeng Zeng, Qitian Wu, and Junchi Yan. 2023. Molerec: Combinatorial drug recommendation with substructure-aware molecular representation learning. In *Proceedings of the ACM Web Conference 2023*. 4075–4085.

[31] Kejing Yin, William K Cheung, Benjamin CM Fung, and Jonathan Poon. 2020. Learning inter-modal correspondence and phenotypes from multi-modal electronic health records. *IEEE Transactions on Knowledge and Data Engineering* 34, 9 (2020), 4328–4341.

[32] Junkun Yuan, Xu Ma, Defang Chen, Fei Wu, Lanfen Lin, and Kun Kuang. 2023. Collaborative semantic aggregation and calibration for federated domain generalization. *IEEE Transactions on Knowledge and Data Engineering* 35, 12 (2023), 12528–12541.

[33] Haijun Zhang, Xian Yang, Liang Bai, and Jiye Liang. 2023. Enhancing Drug Recommendations via Heterogeneous Graph Representation Learning in EHR Networks. *IEEE Transactions on Knowledge and Data Engineering* (2023).

[34] Jize Zhang, Bhavya Kailkhura, and T Yong-Jin Han. 2020. Mix-n-match: Ensemble and compositional methods for uncertainty calibration in deep learning. In *International conference on machine learning*. PMLR, 11117–11128.

[35] Xianli Zhang, Buyue Qian, Yang Li, Shilei Cao, and Ian Davidson. 2021. Context-aware and time-aware attention-based model for disease risk prediction with interpretability. *IEEE Transactions on Knowledge and Data Engineering* 35, 4 (2021), 3551–3562.

[36] Yutao Zhang, Robert Chen, Jie Tang, Walter F Stewart, and Jimeng Sun. 2017. LEAP: learning to prescribe effective and safe treatment combinations for multimorbidity. In *proceedings of the 23rd ACM SIGKDD international conference on knowledge Discovery and data Mining*. 1315–1324.

[37] Zhilu Zhang, Adrian V Dalca, and Mert R Sabuncu. 2019. Confidence calibration for convolutional neural networks using structured dropout. *arXiv preprint arXiv:1906.09551* (2019).


## A Proof of Proposition 4.1

Recall that the Proposition 4.1 is expressed as

Proposition A.1. *Given a backbone and the accuracy-aware binning scheme, there exits* $acc_i - acc_{i+1} \geq 0$. *Furthermore, assume that* $\left|\hat{Y}_{<i}\right| = \beta_i |Y|$, *then* $acc_i - acc_{i+1}$ *has a lower bound of*

$$\frac{(1-\alpha_{i+2}-\beta_{i+1})\alpha_i + (1-\alpha_i-\beta_{i+2})\alpha_{i+1} + (1-\alpha_{i+1}-\beta_i)\alpha_{i+2}}{(\beta_{i+1}-\beta_i)(\beta_{i+2}-\beta_{i+1})}.$$

We begin our proof with the numerical relationship between accuracy and FNR value.

Proof. Given that we have predetermined FNR values as

$$\alpha_i = \mathrm{FNR}(\hat{Y}_{<i}, Y) = 1 - \frac{\left|Y \cap \hat{Y}_{<i}\right|}{|Y|}. \tag{10}$$

**Table 4: Performance comparison between UM and ARC. The data has been disturbed in another way and is supplementary to Table 2.**

| MR model | Method | MIMIC-III | | | MIMIC-IV | | |
|---|---|---|---|---|---|---|---|
| | | ECE (↓, %) | ACC$^+$ (↑, %) | NLL (↓) | ECE (↓, %) | ACC$^+$ (↑, %) | NLL (↓) |
| Carmen | UM | $14.17_{\pm 6.85}$ | $58.38_{\pm 6.12}$ | $0.9259_{\pm 0.1966}$ | $10.74_{\pm 0.38}$ | $55.06_{\pm 0.53}$ | $0.6802_{\pm 0.0138}$ |
| | ARC | $\mathbf{4.68_{\pm 1.93}}$ | $\mathbf{69.42_{\pm 0.26}}$ | $\mathbf{0.5933_{\pm 0.0095}}$ | $\mathbf{4.93_{\pm 5.51}}$ | $\mathbf{67.13_{\pm 5.20}}$ | $\mathbf{0.6535_{\pm 0.1594}}$ |
| SafeDrug | UM | $17.15_{\pm 0.94}$ | $54.05_{\pm 0.46}$ | $0.9491_{\pm 0.05076}$ | $7.20_{\pm 0.29}$ | $68.94_{\pm 0.16}$ | $0.6341_{\pm 0.0058}$ |
| | ARC | $\mathbf{6.66_{\pm 1.04}}$ | $\mathbf{67.50_{\pm 0.34}}$ | $\mathbf{0.6264_{\pm 0.0111}}$ | $\mathbf{5.24_{\pm 0.69}}$ | $\mathbf{68.99_{\pm 0.19}}$ | $\mathbf{0.6117_{\pm 0.0099}}$ |
| GAMENet | UM | $21.97_{\pm 0.76}$ | $46.64_{\pm 0.53}$ | $0.6756_{\pm 0.0033}$ | $20.08_{\pm 1.58}$ | $52.20_{\pm 0.46}$ | $0.9662_{\pm 0.0916}$ |
| | ARC | $\mathbf{9.28_{\pm 2.36}}$ | $\mathbf{67.56_{\pm 0.33}}$ | $\mathbf{0.6489_{\pm 0.0195}}$ | $\mathbf{10.53_{\pm 3.53}}$ | $\mathbf{57.25_{\pm 2.83}}$ | $\mathbf{0.6757_{\pm 0.0515}}$ |

**Table 5: ECE, ACC$^+$, and NLL loss values for the ablation study. ARC w/o CRC denotes ARC without conformal risk control (CRC) and is a variant of ARC.**

| MR Model | Method | MIMIC-III | | | MIMIC-IV | | |
|---|---|---|---|---|---|---|---|
| | | ECE (↓, %) | ACC$^+$ (↑, %) | NLL (↓) | ECE (↓, %) | ACC$^+$ (↑, %) | NLL (↓) |
| Carmen | ARC | $1.63_{\pm 0.55}$ | $\mathbf{69.84_{\pm 1.13}}$ | $\mathbf{0.5693_{\pm 0.0203}}$ | $1.54_{\pm 0.65}$ | $\mathbf{67.66_{\pm 0.50}}$ | $\mathbf{0.6027_{\pm 0.0056}}$ |
| | ARC w/o CRC | $\mathbf{1.54_{\pm 0.44}}$ | $69.68_{\pm 1.25}$ | $0.5718_{\pm 0.0215}$ | $\mathbf{1.51_{\pm 0.68}}$ | $67.61_{\pm 0.53}$ | $0.6039_{\pm 0.0050}$ |
| SafeDrug | ARC | $4.84_{\pm 0.23}$ | $\mathbf{67.54_{\pm 0.76}}$ | $\mathbf{0.6173_{\pm 0.0130}}$ | $\mathbf{1.34_{\pm 0.50}}$ | $\mathbf{70.41_{\pm 0.41}}$ | $\mathbf{0.5538_{\pm 0.0092}}$ |
| | ARC w/o CRC | $\mathbf{4.79_{\pm 0.23}}$ | $67.46_{\pm 0.79}$ | $0.6189_{\pm 0.0132}$ | $1.35_{\pm 0.49}$ | $70.30_{\pm 0.36}$ | $0.5590_{\pm 0.0063}$ |
| GAMENet | ARC | $2.58_{\pm 0.26}$ | $\mathbf{72.67_{\pm 0.32}}$ | $\mathbf{0.5089_{\pm 0.0114}}$ | $3.23_{\pm 0.80}$ | $69.76_{\pm 1.06}$ | $\mathbf{0.5369_{\pm 0.0219}}$ |
| | ARC w/o CRC | $\mathbf{2.48_{\pm 0.22}}$ | $72.54_{\pm 0.37}$ | $0.5152_{\pm 0.0126}$ | $\mathbf{3.04_{\pm 0.86}}$ | $\mathbf{71.92_{\pm 0.29}}$ | $0.5527_{\pm 0.0240}$ |

We can deduce a novel formulation for the bin-wise accuracy $acc_i$ as:

$$acc_i = \frac{\left|Y \cap (\hat{Y}_{<i+1} - \hat{Y}_{<i})\right|}{\left|(\hat{Y}_{<i+1} - \hat{Y}_{<i})\right|} = (\alpha_i - \alpha_{i+1}) \frac{|Y|}{\left|\hat{Y}_{<i+1}\right| - \left|\hat{Y}_{<i}\right|}. \tag{11}$$

Then the accuracy differences between adjacent bins $b_i$ and $b_{i+1}$ can be written as

$$acc_i - acc_{i+1} = \frac{(\alpha_i - \alpha_{i+1})\left|\hat{Y}_{<i+2}\right| + (\alpha_{i+2} - \alpha_i)\left|\hat{Y}_{<i+1}\right| + (\alpha_{i+1} - \alpha_{i+2})\left|\hat{Y}_{<i}\right|}{(\left|\hat{Y}_{<i+1}\right| - \left|\hat{Y}_{<i}\right|)(\left|\hat{Y}_{<i+2}\right| - \left|\hat{Y}_{<i+1}\right|)}. \tag{12}$$

Given $\mathrm{FNR}(\hat{Y}_{<i}, Y) = \alpha_i$, according to Equation (2), there exists $|Y \cap \hat{Y}_{<i}| = (1 - \alpha_i)|Y|$, which is equivalent to $\left|\hat{Y}_{<i}\right| \geq (1 - \alpha_i)\,|Y|$, leading to $(\alpha_i - \alpha_{i+1})\left|\hat{Y}_{<i+2}\right| \geq (\alpha_i - \alpha_{i+1})(1 - \alpha_{i+2})|Y|$. Likewise, we have $(\alpha_{i+2} - \alpha_i)\left|\hat{Y}_{<i+1}\right| \geq (\alpha_{i+2} - \alpha_i)(1 - \alpha_{i+1})|Y|$ and $(\alpha_{i+1} - \alpha_{i+2})\left|\hat{Y}_{<i}\right| \geq (\alpha_{i+1} - \alpha_{i+2})(1 - \alpha_i)|Y|$. With Equation (12), we can deduce that

$$acc_i - acc_{i+1} \geq \frac{[(\alpha_i - \alpha_{i+1})(1 - \alpha_{i+2}) + (\alpha_{i+2} - \alpha_i)(1 - \alpha_{i+1}) + (\alpha_{i+1} - \alpha_{i+2})(1 - \alpha_i)]\,|Y|}{(\left|\hat{Y}_{<i+1}\right| - \left|\hat{Y}_{<i}\right|)(\left|\hat{Y}_{<i+2}\right| - \left|\hat{Y}_{<i+1}\right|)}.$$

We can conclude that

$$acc_i - acc_{i+1} \geq 0. \tag{13}$$

The non-negativity of $acc_i - acc_{i+1}$ proves the robustness of our accuracy-aware binning method.

Now that we have the conclusion that there exists a difference between $acc_i$ and $acc_{i+1}$, we can further calculate the lower bound of this difference as below.

Assume that $\left|\hat{Y}_{<i}\right| = \beta_i|Y|$, we have $(\alpha_i - \alpha_{i+1})\left|\hat{Y}_{<i+2}\right| \geq \alpha_i(1 - \alpha_{i+2})\,|Y| - \alpha_{i+1}\beta_{i+2}|Y|$. Similarly, we have $(\alpha_{i+2} - \alpha_i)\left|\hat{Y}_{<i+1}\right| \geq \alpha_{i+2}(1 - \alpha_{i+1})\,|Y| - \alpha_i\beta_{i+1}|Y|$, $(\alpha_{i+1} - \alpha_{i+2})\left|\hat{Y}_{<i}\right| \geq \alpha_{i+1}(1 - \alpha_i)\,|Y| - \alpha_{i+2}\beta_i|Y|$. Then,

$$acc_i - acc_{i+1} \geq \frac{(1 - \alpha_{i+2} - \beta_{i+1})\alpha_i + (1 - \alpha_i - \beta_{i+2})\alpha_{i+1} + (1 - \alpha_{i+1} - \beta_i)\alpha_{i+2}}{(\beta_{i+1} - \beta_i)(\beta_{i+2} - \beta_{i+1})}. \tag{14}$$

With the lower bound of the accuracy differences between adjacent bins, i.e., $acc_i - acc_{i+1}$, our accuracy-aware binning scheme can ensure the numerical diversity of $\{acc_i\}_{i=1}^{B}$. □

# B Additional Results

## B.1 More Results of the Disturbed Data

Recall that to test the robustness of our method, the comparisons with all the baselines are presented in Table 2. Overall, it can be observed that ARC consistently shows superiority.

Notably, among other methods, the performance of UM is drastically reduced compared to IR and FW. This is because our disturbance sets a threshold for raw confidence, and UM focuses on the number of samples and neglects the value of raw confidence. The robustness of ARC demonstrates the theoretical analysis that our accuracy-aware binning scheme can split the data more properly, thus providing a more reliable recommendation. For the sake of fairness, we change the way of disturbance by eliminating the label of the samples with the Top-1000 raw confidences, *i.e.*, disturbing

the distribution based on the number of samples instead of the value of raw confidences. We show the comparison between ARC and UM in Table 4. It can be observed that ARC still consistently outperforms UM on all metrics. This observation further establishes the efficacy and robustness of our method.

## B.2 Ablation Study

We conducted the ablation study to validate the efficacy of accuracy-aware and CRC-based binning schemes respectively. The results of ***ECE***, ***ACC***$^+$, and ***NLL*** loss are presented in Table 5, where ARC w/o CRC denotes ARC without conformal risk control (CRC), i.e., only applies the accuracy-aware binning scheme. It is evident that ARC consistently achieves the lowest ***NLL*** loss in comparison to ARC w/o CRC. This is attributed to the adaptability of ARC in considering the calibration data size when determining bin edges for test data via conformal risk control. Despite the slightly superior performance of ARC w/o CRC on ***ECE***, safety-critical clinical tasks would prefer ARC due to its better performance in terms of ***ACC***$^+$ and ***NLL***. It is noteworthy that even by solely employing the theoretically guaranteed accuracy-aware binning scheme, our method still achieves the best performance compared to the contents in Table 2. To sum up, ARC achieves more discernible confidence calibration by both accuracy-aware binning and CRC-based binning schemes.